

me, these risk metrics are acceptable, but then again living on \$20,000 per year (that is, a 2% spending rate) will be difficult.

Here is another numerical example of the sustainability probability for more realistic portfolio and retirement values. Say you retire with \$1,000,000 at the age of 65 when your median remaining lifespan is approximately 19 years. You plan to withdraw \$40,000 per year adjusted for your personal inflation rate, which is, thus, a 4% spending rate. Also, assume that your million dollar nest egg is invested in an equity-based mutual fund that is expected to earn an optimistic 7% after (your personal) inflation rate and the estimated volatility rate is 18%. Using values of $S=0.04$, $MRL=19$, $AM=0.07$, and $VOL=0.18$, according to the equation in Figure 8.3, your retirement alpha is 3.15 units and your retirement beta is 1.16 units. Finally, going down the columns and across the rows, the relevant entry in Table 8.3 is slightly greater than 90%. To me, a 90% sustainability rate is marginally acceptable and is often called the “prudent 4% rule” of retirement spending.

Note that higher values of alpha are good for your retirement, as are lower values of beta. As you move to the lower-right corner, the sustainability numbers decrease, and as you move to the upper-left corner, the sustainability numbers increase. Intuitively, this should make sense if you look carefully at the definitions of alpha and beta. Notice that higher spending rates increase your retirement beta (not good), while higher portfolio returns increase your retirement alpha (good). Also, greater investment volatility reduces your retirement alpha (not good) but also reduces your retirement beta.

Now that you have an intuitive feel for the table, you might wonder where exactly the numbers came from. Or how do you generate your own table with different values of alpha and beta? Actually, this is where the formula I mentioned earlier is used. The tabular values, which are not based on any Monte Carlo simulations, can be obtained by typing the expression `=1-GAMMADIST(beta,alpha,1,TRUE)` within a cell of any Microsoft Excel spreadsheet. In place of the words *alpha* and *beta*, use their actual numerical values calculated using the Figure 8.3 formulas and out pops the precise number from the table.

If you want proof, check out some of the references at the end of the book. And for those of you who remain skeptical that a simple analytic formula offers a shortcut around cumbersome and expensive Monte Carlo simulations of retirement income, I urge you to compare